HW 3

(1) min: $x_1 - x_2 + 0 \cdot x_3 + 0 \cdot x_4$

s.t. $2x_1 + 3x_2 - x_3 + x_4 \leq 0$

$3x_1 + x_2 + 4x_3 - 2x_4 \geq 3$

$-x_1 - x_2 + 2x_3 + x_4 = 6$

$x_1 \leq 0 \quad x_2, x_3 \geq 0$

$$\min_{x_1 \leq 0,\ x_2, x_3 \geq 0} \max_{p_1 \leq 0,\ p_2 \geq 0,\ p_3} : x_1 - x_2 + p_1(0 - (2x_1 + 3x_2 - x_3 + x_4)) + p_2(3 - (3x_1 + x_2 + 4x_3 - 2x_4)) + p_3(6 - (-x_1 - x_2 + 2x_3 + x_4))$$

$$\max_{p_1 \leq 0,\ p_2 \geq 0,\ p_3} \min_{x_1 < 0,\ x_2, x_3 \geq 0} (1 - 2p_1 - 3p_2 + p_3)x_1 + (-1 - 3p_1 - p_2 + p_3)x_2 + (0 + p_1 - 4p_2 - 2p_3)x_3 + (0 - p_1 - 2p_2 - p_3)x_4 + 3p_2 + 6p_3$$

Rule: $(b - Ax) \geq 0$ if $(b - Ax) \leq 0$ Penalty $p \leq 0$

$(b - Ax) \leq 0$ if $(b - Ax) \geq 0$ Penalty $p \geq 0$

$$\max_{p_1 \leq 0,\ p_2 \geq 0,\ p_3} 3p_2 + 6p_3$$

s.t. $(2p_1 + 3p_2 - p_3) \leq 1$

$3p_1 + p_2 - p_3 \geq -1$

$-p_1 + 4p_2 + 2p_3 \geq 0$

$p_1 + 2p_2 + p_3 = 0$

2(ii) Let $Ax \geq 0$ and $x \geq 0$ and $x_1 = 0$

Implies $Ax \geq 0$, $x \geq 0$ and $x_1 = 1$

or, any $Ax \geq 0$ $x \geq 0$ and $x_1 > 0$ is NOT feasible.

$A \in R^{m \times n}$, $x \in R^n$, $Ax \in R^m$

<u>Farkas Lemma</u>:- There will be a seperating Hyperplane when $\exists d$, s.t.

$d^T a_i \geq 0$ for all i, but $d^T c < 0$

Extending this

Let there be a vector $P \geq 0$, s.t. ; $P \in R^m$

$P^T A \leq 0$ and $P^T A_1 < 0$

(the first element first column..)

we have $Ax = \sum_{i=1}^{n} A_i x_i$

$$P^T A x = \sum_{i=1}^{n} x_i (P^T A_i) = \text{scalar}$$

$P^T A \leq 0$, $P^T A_1 < 0 \Rightarrow$

and if $x_1 \neq 0$ and $x_1 > 0$

$$P^T A x < 0$$

This is a contradiction

This shows

Hence / $Ax \geq 0$ and $x \geq 0$
and $x_1 \neq 0$ is not feasible

Hence it implies

$$Ax \geq 0, x \geq 0 \text{ and } x_1 = 0$$

(ii)

Let there exists $P \geq 0$ S.T

$$P^T A \leq 0 \text{ and } P^T A_1 < 0$$

For $x \geq 0$, $Ax \geq 0$

$$P^T A \leq 0$$

Given $x \geq 0$ and $P^T A \leq 0$ $\Rightarrow$ $(P^T A) a \leq 0$

$$P^T (Ax) \leq 0$$

However $P \geq 0$ and $(Ax) \geq 0$

The first column $P^T A_1 < 0$, but $x \geq 0$

This can be satisfied iff $x_1 = 0$.

$\therefore$ It follows $Ax \geq 0$, $x \geq 0$ and $x_1 = 0$

(3) Boyd 5.2

$$\left.\begin{array}{l} Ax \le b \\ a_i^T x \le b \end{array}\right] \text{ is } \underline{\text{infeasible}}. \quad \text{— (1)}$$

is a system of 500 eq, and 100 variables

$A \in R^{500}$, $x \in R^{100}$

~~The dual of the system~~

Let the above be represented as an LP:

$$\min \; k \quad (\text{when } \underline{k=0})$$
$$\text{s.t. } Ax \le b.$$

The dual can be written as

$$\min \max_{P \ge 0} \; 0 + (b - Ax)P$$

$$\max_{P \ge 0} \min \; bP - APx$$

$\longrightarrow x$ is unbounded, so $AP \Rightarrow \emptyset$

$$\left.\begin{array}{ll} \max & p^T b \\ P \ge 0 & \text{s.t. } A^T P = 0 \\ & P \ge 0 \end{array}\right] \text{ — (2) Feasible per Farkas}$$

$\boxed{P \in R^{500}}$, $A \in R^{500 \times 100}$

Given that $Ax \le b$ is infeasible per Farkas theorem (2) is possible

The dual is $A^T P$ is a system

(3) of 100 equations, ~~and 500 unknown~~

and 500 unknowns $(P_1, P_2 \ldots P_{500})$

Per Farkas' Theorem as

$$\max_{P \geq 0} \quad P^T b \quad \text{s.t.} \quad \left.\begin{array}{l} A^T P = 0 \\ P \geq 0 \end{array}\right] \text{Exists}$$

in a system of 100 equations)

Per Linear Algebra, basic feasibility - a feasible solution to dual - can be expressed as a combination of 100+1 = 101 P vectors (points).

[This implies that a subset of at most 101 inequalities in Primal are infeasible